EXERCISE

1. Encircle the correct choice

i) A self-regulating process by which biological systems maintain stability while adjusting to the changing conditions called
(a) Homeostasis (b) Osmoregulation
(c) Excretion (d) Biological rhythms

ii) Which does not maintain a stable, homeostatic condition rather it intensifies the change that is happening to the body.
(a) Negative feedback (b) Positive feedback
(c) Feed back system (d) Excretion

iii) Fresh water organisms also have specialized cells located in their gills and in skin which actively extract Na^+, Cl^- and Ca^+ from external medium and excrete (H^+) or basic (HCO_3-) for acid base balance in the body fluids are called
(a) Granulocyte (b) Lymphocyte
(c) Ionocytes (d) podocyte

iv) Renal cortex produces an important hormone necessary for the synthesis of RBC's
(a) Erythropoietin (b) Leukopoietin
(c) Thrombopoietin (d) Renin

v) Glomerulus have specialized cells that are wrapped around blood capillaries that play an active role in preventing plasma proteins from entering the urinary ultrafiltrate called
(a) Epithelial cells (b) Podocyte cells
(c) Endothelial cells (d) None of them

vi) Tubules secretes ions such as hydrogen, potassium, and NH_3 into the filtrate while reabsorbing the HCO^{3-} from the filtrate are called
(a) Distal convoluted tubule (b) Proximal convoluted tubule
(c) Collecting duct (d) Loop of Henle

vii) Glomerular filtration rate (GFR) is in which proportional to the hydrostatic pressure exerted in glomerulus wall
(a) Indirectly proportional (b) Directly proportional
(c) Same proportional (d) High proportional

viii) Another compound that increases the osmotic gradient in inner medulla is the
(a) Urea (b) Water
(c) Sulphates (d) Phosphates

ix) The inflammation in kidneys due to irritation of kidney stones is called
(a) Lithonephritis (b) Edema
(c) Sarcoma (d) Encephalitis

x) If the surrounding environment becomes hot or a person do strenuous physical activity the heat produced inside the body and raises body temperature above suitable limit this condition is called
(a) Exothermia (b) Hyperthermia
(c) Hypothermia (d) Endothermia

2. Write short answers of the following questions:

i) Why the physiological integration of internal body environment is important for living organisms?
ii) Why it is important that positive and negative feedback mechanisms counter act upon each other?
iii) How the aquatic Osmoregulators overcome the osmoregulatory problems?
iv) Why animals excrete different types of excretory waste?
v) What is the use of counter current mechanism in kidney?
vi) What is the role of kidney as an endocrine organ?
vii) How kidney stones are formed?
viii) What is the impact over the human body if kidneys suddenly or gradually loss the filtering abilities?
ix) What are the problems associated with kidney transplant?
x) What is hypothermia? How body overcome this condition?

3. Write detailed answers of the following questions:

i) Explain the detailed structure of nephron with labelled diagram
ii) Explain the functions of kidney
iii) Explain the causes of kidney failure and its treatment

EXERCISE

1. Encircle the correct choice.

i) The type of bone cell that tears down bone during the building and remodeling process are
(a) Osteocytes (b) Osteoblasts
(c) Osteoclasts (d) Bone lining cells

ii) During bone formation, as the periosteum calcifies, it gives rise to a thin plate of compact bone called
(a) Periosteal bud (b) Periosteal bone collar
(c) Primary ossification center (d) Epiphyseal plate

iii) The ring like shape of the girdle is due to the joining of coxal bone with the sacrum of vertebral column at anterior side and below by a joint in between pubic part called
(a) Pubic symphysis (b) Iliac junction
(c) Pelvic symphysis (d) Syndesmosis

iv) Cartilage is a soft flexible form of connective tissue surrounded by a layer called
(a) Myocardium (b) periosteum
(c) Osteoderm (d) Perichondrium.

v) An antagonist muscle is relaxed while the other side opposite to antagonist muscle
(a) Contracted (b) Relaxed
(c) No change (d) Extremely relaxed

vi) The stage when old bone tissues are removed and gets replaced with new ones that joins both fractured sides called
(a) Bone remodeling stage (b) Bone callus formation
(c) Cartilaginous callus formation (d) Hematoma formation

vii) Which is related to the mild to severe grade of stretching or tearing of a ligament that holds the bones of a joint together?
(a) Spasm (b) Tetany
(c) Sprains (d) Cramps

viii) Muscle fiber are gathered in many groups each group of fibers is called a fascicle which is surrounded by a membrane called
(a) Sarcolemma (b) Perimysium
(c) Endomysium (d) Epimysium

ix) The heart muscles arranged in a branching network where cells are joined together and making a
(a) Syncytium (b)Sarcoplasmic reticulum
(c) Endoplasmic reticulum (d) T-tubules

x) These joints hold the bone by dense connective tissue contain collagenous fibers
(a) Fibrous joints (b) Syndesmosis
(c) Gomphosis (d) Synovial joints

2. Write short answers of the following questions:

1. How is the cartilage useful for our skeletal system?
2. Give a comparison of major types of joints
3. How is the healing of fractured bone initiated pain developing?
4. How to conduct a first aid treatment for joint injuries and fractures?
5. Give a comparison of types of muscles.
6. Why is calcium essential for cross bridge formation?
7. How the nervous coordination regulates muscle contraction mechanism?
8. How is tetany different from tetanus?
9. Explain the different types of cartilage.

3. Write detail answers of the following questions:

1. Explain the structure of skeletal muscles and ultra structure of skeletal muscle fiber with labelled diagram.
2. Explain the repair process of simple fractures
3. Explain the cross-bridge cycle of muscle contraction with labelled diagram

- The CNS is protected in three ways the first line of defense is bony armor, beneath bony armor a triple layer of connective tissue called meninges and cerebrospinal fluid cushion the brain and spinal cord.
- The forebrain is the largest and most obvious part of human's brain.
- The midbrain is extremely reduced in humans but an important relay center.
- Hind brain consists of medulla, pons and cerebellum.
- The PNS consists of paired cranial and spinal nerves and associated ganglia.
- Sense of smell involves chemoreceptors located in the upper portion of the nasal cavity.
- The receptors cells for taste are modified epithelial cells organized into taste buds.
- Drug that provide pain relief, such as morphine or Demerol, block synaspses in the pain pathways of the brain or spinal cord.

EXERCISE

1. Encircle the correct choice.

i) Most of the neurons in the human brain are

(a) Sensory neurons (b) Motor neurons
(c) Interneurons (d) Auditory neurons

ii) A nervous system can alter activities in its target cells in muscles and gland because

(a) They are electrically coupled by gap junctions.
(b) The target cells have receptor proteins for the signals released by the nervous system.
(c) The nervous system releases signals into the blood to control the target cells.
(d) The target cells that become disconnected from the nervous system rapidly die.

iii) For a neuron with an initial membrane potential at -70mv, an increase in the movement of potassium ions out of that neuron's cytoplasm would result in
(a) Depolarization of the neuron
(b) Hyperpolarization of neuron
(c) The replacement of potassium ions with sodium ions.
(d) The replacement of potassium ions with calcium ions.

iv) Action potentials move along axons
(a) More slowly in axons of large than in small diameter.
(b) The direct action of acetylcholine on the axon membrane.
(c) Activating the sodium-potassium pump at each point along the axon membrane.
(d) More rapidly in myelinated than in non-myelinated axons.

v) The surface on a neuron that discharges vesicles is the
(a) Dendrite (b) Axon hillock
(c) Presynaptic membrane (d) Postsynaptic membrane.

vi) An inhibitory postsynaptic potential occurs in a membrane made more permeable to
(a) Potassium ions (b) Sodium ions
(c) Calcium ions (d) ATP

vii) The major inhibitory neurotransmitter of the brain is
(a) Acetylcholine (b) Epinephrine
(c) GABA (d) Endorphin

viii) Where are neurotransmitter receptors located
(a) On the nuclear membrane
(b) In the myelin sheet
(c) At nodes of Ranvier
(d) On the postsynaptic membrane

ix) Which selection is incorrectly paired?
(a) Forebrain → Diencephalon
(b) Fore brain → Cerebrum
(c) Midbrain → Brainstem
(d) Midbrain → Cerebellum

x) Which of the following receptors is incorrectly paired with the type of energy is transduces?
(a) Mechanoreceptors → Sound
(b) Pain receptors → Electricity
(c) Chemoreceptors → Solute concentration
(d) Thermo receptors → Heat

2. Write short answers of the following questions:

i) Why receptors are called transducers?
ii) Why neurolemma is polarized?
iii) What do you mean by threshold potential?
iv) Why impulses move faster in myelinated neurons?
v) Why interneurons are called association neuron?
vi) What do you mean by saltatory conduction?
vii) Differentiate between following
(a) Electrical and Chemical synapse
(b) CNS and PNS
(c) Excitatory neurotransmitters and Inhibitory neurotransmitters
(d) Parasympathetic nervous system and Sympathetic nervous system
viii) Define following terms
(a) Endorphins
(b) Pacinian corpuslces
(c) Nociceptors
(d) Reflex arc
(e) Stroke
(f) Meningitis

3. Write detailed answers of the following questions:

i. Describe structure and function of neurons.
ii. Describe mechanism of action membrane potential.
iii. Explain vascular and infectious disorders of CNS.
iv. Describe anatomy and physiology of human brain.
v. Describe sensory receptors and their working.

- Ovaries are female gonads produce both egg and female sex hormones.
- Placenta an endocrine tissue release variety of hormones during pregnancy like HCG and progesterone.

EXERCISE

1. Encircle the correct choice.

i) Which of the following statements about hormones is incorrect?
(a) They are produced by endocrine glands.
(b) They are modified amino acids, peptides or steroid molecules.
(c) They are carried by the circulatory system.
(d) They are used to communicate between different organisms.

ii) Oxytocin and ADH are produced by the _____ and stored in the ____.
(a) Hypothalamus; Neurohypophysis
(b) Adenohypophysis; Kidneys
(c) Adrenal cortex; Adrenal medulla
(d) Posterior pituitary; Anterior pituitary

iii) Tropic hormones from the anterior pituitary directly affect the release of which of the following?
(a) Parathyroid hormone (b) Calcitonin
(c) Epinephrine (d) Thyroxine

iv) Which of the following endocrine disorders is /are not correctly matched with the malfunctioning gland?
(a) Diabetes → Pancreas (b) Giantism → Pituitary
(c) Goiter → Adrenal medulla (d) Tetany → Parathyroid

v) Which hormone exerts antagonistic action to PTH?
(a) Thyroxin (b) Calcitonin
(c) Growth hormone (d) Epinephrine

vi) Which of the following are synthesized from the amino acid tyrosine?
(a) Epinephrine (b) Catecholamines
(c) Thyroxin (d) All of the following

vii) If the adrenal cortex were removed, which group of hormones would be most affected?
(a) Steroid (b) Peptide
(c) Tropic (d) Amino acid derived

viii) Which of the following hormones is (are) secreted by the adrenal gland in response to stress and promote(s) the synthesis of glucose from non-carbohydrate substrates?
(a) Glucagon (b) Glucocorticoids
(c) Epinephrine (d) Thyroxin

ix) Which of the following stimulates the contraction of uterine muscle?
(a) Oxytocin (b) Thyroxin
(c) Growth hormone (d) Insulin

x) A distinctive feature of the mechanism of action of thyroid hormones and steroid hormones is that
(a) These hormones are regulated by feedback loop
(b) These hormones bind with specific receptor protein on the plasma membrane of target cells.
(c) These hormones bind to receptors inside cells
(d) Target cells react more rapidly to these hormones than to local regulators.

2. Write short answers of the following questions:

i) Why hydrophilic hormones need secondary messengers?
ii) Why hormones are called chemical messengers?
iii) Why posterior pituitary is called neurohypophysis?
iv) Enlist hormones of anterior pituitary?
v) Why pancreas is called as exocrine and endocrine gland?
vi) Why secretions of adrenal cortex are called corticosteroids?
vii) Differentiate between following
(a) Hypothyroidism and Hyperthyroidism
(b) Insulin and glucagon
(c) Calcitonin and Parathyroid hormone
(d) Lipophilic and hydrophilic hormones
viii) Define following terms
(a) Hormones (b) Goiter (c) Thymosin
(d) GnTH (e) Atrial natriuretic hormone (f) Erythropoietin

EXERCISE

1. Encircle the correct choice.

i) The decrease in response to repeated or continuous stimulation is called
(a) Insight (b) Habituation
(c) Maturation (d) Instinct

ii) The cyclic physiological patterns of activities in an organism that are in response to periodic environmental changes is called
(a) Biological Clock (b) Biological rhythms
(c) Circadian rhythms (d) All of Above

iii) A directional movement of an organism toward or away from the particular stimulus is called
(a) Kinesis (b) Taxis
(c) Reflex (d) Fixed Action Pattern

iv) Simplest form of learning, in which the animals learn not to respond or ignore to a repeated, irrelevant stimulus.
(a) Imprinting (b) Kinesis
(c) Habituation (d) Fixed Action Pattern

v) Which one of the following is not an instinct behaviour?
(a) Migration of Salmon Fish (b) Dances of bees
(c) Territorial behaviour in Gorillas (d) Nest building by birds

vi) Wolfgang Kohler in 1920s performed the experiment on the behaviour of
(a) Rats (b) Chimpanzees
(c) Graylag Goose (d) Dog

vii) Workers (honey bee) are
(a) Fertile Females (b) Fertile Male
(c) Sterile Female (d) Sterile Male

viii) Agonistic behaviour is the type of
(a) Innate Behaviour (b) Learning behaviour
(c) Social Behaviour (d) Instinctive behaviour

ix) Migration of Salmon Fish is the example of
(a) Orientation behaviour (b) Learning behaviour
(c) Social Behaviour (d) Instinctive behaviour

x) Skinner box was used for
(a) Classic Conditioning (b) Operant Learning
(c) Latent Learning (d) Insight Learning

EXERCISE

1. Encircle the correct choice.

i) Ovarian cycle includes changes in ovarian follicles to secondary follicle and finally to
(a) Graafian follicle (b) Primary follicle
(c) Vesicular follicle (d) Both “a” and “c”

ii) In males, FSH promotes the development and maturation of?
(a) Testosterone (b) Sperm cells
(c) Egg cells (d) Polar cells

iii) The menstrual cycle is primarily regulated by all of the following except:
(a) FSH (b) Estrogen
(c) Androgen (d) Progesterone

iv) FSH induces the Graafian follicle to produce
(a) LH (b) Estrogen
(c) Androgen (d) Progesterone

v) Key hormones in female reproductive system that regulates menstrual cycle.
I. Estrogen II. Progesterone III. Testosterone
(a) I only (b) II only
(c) I and II (d) III only

vi) What is the medical term for a miscarriage?
(a) Spontaneous abortion (b) Induced abortion
(c) Ectopic pregnancy (d) Both “a” and “b”

vii) Syphilis is a sexually transmitted disease caused by
(a) Bacteria (b) Virus
(c) Protozoan (d) Yeast

viii) Structure that protects the testes and keeps them at a temperature several degrees below the normal body temperature.
(a) Epididymis (b) Scrotum
(c) Penis (d) All of the above

ix) In-vitro fertilization (IVF) is a method used to:
(a) Treat infertility (b) Enhance hormones
(c) Prevent STDs (d) None of the above

x) Which of the following is a potential cause of female infertility?
(a) Polycystic ovary syndrome (PCOS) (b) Endometriosis
(c) Fallopian tube blockage (d) All of these

EXERCISE

1. Encircle the correct choice

i) The genetic material of a fertilized egg is a combination of
(a) The mother's DNA only
(b) The father's DNA only
(c) Both the mother's and father's DNA
(d) None of the above

ii) Human embryonic development occurs in the
(a) Uterus (b) Fallopian tube
(c) Cervix (d) Vagina

iii) The process of cell differentiation involves:
(a) The formation of gametes
(b) The formation of zygotes
(c) The specialization of cells for different functions
(d) The replication of DNA

iv) The implantation of the embryo in the uterus occurs at
(a) 1 week after fertilization (b) 2 weeks after fertilization
(c) 4 weeks after fertilization (d) 8 weeks after fertilization

v) Aging is characterized by:
(a) The accumulation of genetic mutations
(b) The breakdown of cellular processes
(c) The cessation of mitosis
(d) The reduction in DNA replication

vi) The inner cell mass of a blastocyst gives rise to:
(a) The placenta (b) The embryo
(c) The umbilical cord (d) The amniotic fluid

vii) The process of fertilization occurs in the
(a) Uterus (b) Fallopian tube
(c) Cervix (d) Vagina Answer

viii) The stages of embryonic development are
(a) Cleavage, implantation, gastrulation, and organogenesis
(b) Fertilization, implantation, gastrulation, and organogenesis
(c) Cleavage, fertilization, gastrulation, and organogenesis
(d) Cleavage, fertilization, implantation, and organogenesis

ix) The placenta is responsible for
(a) Providing oxygen and nutrients to the developing embryo
(b) Removing waste products from the developing embryo
(c) Producing hormones that maintain the pregnancy
(d) All of the above

x) The formation of the neural tube occurs during
(a) Cleavage (b) Gastrulation
(c) Organogenesis (d) Implantation

2. Write short answer of the following:

i) Why cleavage in birds and reptiles called meroblastic?
ii) Why morula is named so?
iii) How embryonic tissues influence other embryonic tissues?
iv) Define organizers and differentiate between primary and secondary induction.
v) Differentiate between blastula and gastrula.
vi) Why monozygotic twins having same sex?

3. Write detailed answers to the following questions.

i) Explain blastula/blastocyst with emphasis on segmentation cavity.
ii) List the tissues and organs formed from the three germ layers.
iii) Through experimental narration, describe the role of the nucleus and cytoplasm in controlling development.
iv) Describe the events of development in humans in terms of first, second and third trimesters.
v) Describe the maternal derived abnormalities (rubella, abnormal neural tube, thyroid gland and limb development).

EXERCISE

1. Encircle the correct answer:

i) W. Sutton and Theodor observed this behavior of chromosomes.
(a) Formation (b) Distribution
(c) Staining (d) Replication

ii) Which is not the part of chromosomal theory of inheritance.
(a) Gametes do not make equal hereditary contribution
(b) Chromosome segregate during meiosis
(c) Nucleus is the room of hereditary material
(d) Gametes have one copy of homologous chromosome

iii) Newly formed chromosome has two.
(a) Chromatids (b) Heterologous
(c) Centromere (c) Pairs

iv) The complex of histone octamer and two loops of duplex DNA is.
(a) Chromomer (b) Chromatin
(c) Nucleosome (d) Heterochromatin

v) Small part of DNA which has information to synthesize specific polypeptide chain is.
(a) Genome (b) Locus
(c) Gene (d) Nucleotide

vi) According to replication model, the parental double helix remain intact and conserved called.
(a) Semi conservative (b) Conservative
(c) Disposed (d) Eukaryotic replication

vii) A protein which prevents the re-binding of complementary strand during replication at fork is.
(a) DNA helicase (b) Ligase
(c) SSB (d) Primase

viii) The DNA strand which continuously replicate in the direction of 5' → 3' called.
(a) Lagging strand (b) Leading strand
(c) Primer (d) Okazaki fragment

ix) Process where information present on gene is used to produce a functional product by living organism called.
(a) Transcription
(b) Translation
(c) Gene expression
(d) Regulation of gene expression

x) The set of rules used to store the genetic information within a DNA for particular protein synthesis is.
(a) Gene (b) Genetic codes
(c) Codon (d) Anticodon

2. Write short answer of the following:

i) Why DNA is negatively charged molecules?
ii) Why replication is called semi-conservative process?
iii) Do genes of prokaryotic cell and eukaryotic cells are different in structure? If so give the main differences between them?
iv) What are leading and lagging strand of DNA?
v) Give the name of enzymes involved in replication of DNA, also give their brief functions.
vi) Do introns are transcribed, and involve in translation?
vii) Give changes occur in mRNA during transport from nucleus to cytoplasm.
viii) Give site at ribosome and their functions during translation.
ix) Draw structure of tRNA and give functions of their different sites.
x) Do mutation is always harmful? Justify you answer.

3. Give detail answer of following questions:

i) Describe the chromosomal theory of inheritance?
ii) Describe the semi conservative process of DNA replication?
iii) Describe the chemical structure of chromosome.
iv) Describe the process of transcription during gene expression.
v) What is gene regulation how gene regulate during expression?

- Process of exchange of chromosomal segment between non-sister chromatids of homologous chromosome is called crossing over.
- Chromosome which determine sex of the individual called sex-chromosome.
- Chromosomes which are similar in male and female called Autosomes.
- The alleles of holandric gene are only located at Y-chromosome.
- The Y-linked genes are hemizygous, only located on non-homologous region of Y-chromosome.

EXERCISE

1. Encircle the correct answer:

i) Which of the following would cause phenotypic variations among organisms of the same genotype?
(a) Continuous variation within the species
(b) Different varieties of the same species
(c) Different sexes
(d) Exposure to different environments

ii) In which of the following examples of human inheritance is the inheritance pattern explained by multiple alleles on an autosomal chromosome?
(a) The ABO blood group system (b) Cystic fibrosis
(c) Down's syndrome (d) Haemophilia

iii) What are the phenotypes of the parents of a colour-blind son and a non-carrier daughter with normal colour vision?

	Father	Mother
(a)	Carrier	Normal
(b)	Colour-blind	Carrier
(c)	Colour-blind	Colour-blind
(d)	Normal	Carrier

iv) In the F_2 generation of a dihybrid cross, the phenotypes occurred in the ratio 3:1 what does this result indicate?
(a) The alleles were segregating independently
(b) Polygenic inheritance was involved
(c) Codominance was being shown
(d) The gene loci were linked

v) Two parents, both of blood group A, have a daughter of blood group O. What is the probability that their next child will be a boy who has blood group O?
(a) 0.25 (b) 0.5
(c) 0.75 (d) 1

vi) A boy is colour blind which could be the genotype of his mother?
(a) X^NX^N (b) X^NX^n
(c) X^NY (d) X^nY

vii) Number of chromosome in grass hopper female is 24. How many chromosomes are present in grass hopper male?
(a) 26 . (b) 25.
(c) 24 (d) 23

viii) The allele of holandric gene is located at?
(a) X-Chromosome (b) Y-Chromosome
(c) Autosome (d) None of these

ix) The gene which interferes and masks the phenotype of the phenotype of the non-locus gene called?
(a) Mutant gene (b) Epistatic gene
(c) Pleiotropy (d) Sex-linked

x) x. International society of blood transfusion has found more than?
(a) 10 blood system (b) 20 blood system
(c) 30 blood system (d) 40 blood system

2. Write short answer of the following:

i) Why Rh-incompatibility could be a danger to the developing foetus and mother?
ii) Why haemophilia is common in human male?
iii) What do you mean by sex-influenced trait?
iv) Differentiate between following
   a) Linkage and crossing over
   b) Monohybrid Cross and Dihybrid Cross
   c) X-linked trait and Y-linked trait
   d) Autosomes and sex-chromosome
   e) Incomplete dominance and Co-dominance
v) Why human male are heterogametic?
vi) What do you mean by dichromacy and monochromacy?

- In Founder effect, a new small population separated from the larger one due to some geographical or physical barriers starts reproducing within itself.
- Speciation is the biological process of formation of new species of living organisms.

EXERCISE

1. Encircle the most appropriate response.

i) Which of the following was supporting the theory of special creation?
(a) George Buffon (b) Lamarck
(c) Darwin (d) Suárez

ii) Archaeopteryx is a connecting link between
(a) Amphibia and reptiles (b) Reptiles and Aves
(c) Aves and mammals (d) Fish and amphibia

iii) All of the following are homologous organs except
(a) Wings of bat (b) Wings of butterfly
(c) Wings of bird (d) Flippers of dolphin

iv) Which of the following are analogous organs except:
(a) Wings of insect (b) Wings of bird
(c) Wings of bat (d) Arm of man

v) Life originated in
(a) Air (b) Water
(c) Land (d) Space

vi) *Philosophie Zoologique* was written by
(a) Wallace (b) Darwin
(c) Weismann (d) Lamarck

vii) Which of the following is incorrect regarding Lamarck's theory of Inheritance of acquired characters?
(a) Use and disuse of the organs inherited
(b) Continuously using some organ results in its further strengthening in offspring
(c) Continuously disuse of some organ results in its weakening in offspring
(d) Concrete support to the theory through experiments

viii) The theory on human population which inspired Darwin was proposed by
(a) Malthus (b) Lyell
(c) Hutton (d) Wallace

ix) Theory of Natural Selection was lacking any support from
(a) Biogeography (b) Genetics
(c) Comparative anatomy (d) Molecular Biology

x) Which of the following cannot be the disruptive force for Hardy-Weinberg Theorem?
(a) Selection (b) Non-random mating
(c) Mutation (d) Random mating

2. Write short answers of the following questions.

i) Differentiate between the following:
a) Special Creation and Evolution
b) Lamarckism and Darwinism
c) Invagination theory and Endosymbiotic theory
d) Bottle-neck effect and Founder's effect
e) Convergent and Divergent evolution

ii) How the homologous organs support the theory of evolution?
iii) What are analogous organs? Give example.
iv) How Alfred Wallace contributed Charles Darwin regarding the Natural Selection?
v) What do understand by the "descent with modification"?
vi) How Neo-Darwinism differs from Darwinism?
vii) How the sympatric speciation differs from parapatric speciation?
viii) Justify Lamarck as an early proponent of evolution.
ix) What is genetic drift?
x) List out the factors affecting allele frequency

3. Write detailed answers to the following questions.

i) State and explain the contribution of Lamarck in organic evolution.
ii) Discuss the process of origin of single cell Eukaryotes.
iii) How biogeography and paleontology provide evidences in support of evolution?
iv) What is speciation? Explain different ways of speciation.
v) Describe the Theory of Natural Selection.

EXERCISE

1. Encircle the correct choice.

i) Which one of these mainly causes the greenhouse effect?
(a) Acid rain (b) Ozone layer depletion
(c) Global warming (d) Forest fire

ii) Factors that affect the flow of energy at the trophic level are
(a) Non-predatory deaths (b) Growth & Reproduction
(c) Heat loss (d) Sunlight

iii) The main cause of the increase in the amount of CO_2 in the Earth's atmosphere is
(a) Rapidly growing human population
(b) Burning of a large number of fossil fuels
(c) Increase in the number of industries
(d) Decrease in the forested resources.

iv) All is positive for which of the following processes?
(a) Oxidation of Nitrogen
(b) Melting of ice and evaporation of water
(c) Oxidation of gold
(d) Burning of chlorine

v) Which of the following dietary requirements is involved in nitrogen balance?
(a) Carbohydrates (b) Vitamins
(c) Proteins (d) Essential fatty acids

vi) Which of the following cycle is not a gaseous type of cycle?
(a) Carbon cycle (b) Nitrogen cycle
(c) Phosphorus cycle (d) Oxygen cycle

vii) The source of carbon to plants in the carbon cycle is
(a) Carbonate rocks (b) atmospheric carbon dioxide
(c) Fossil fuels (d) all of the above

viii) Conversion of nitrates to nitrogen is called
(a) Nitrification (b) Denitrification
(c) Ammonification (d) Nitrogen fixation

ix) what is the only source of energy for all ecosystems on Earth?

(a) Water (b) Sun

(c) Animals (d) Plants

x) The pioneers in xerarch succession are the

(a) Foliose lichens (b) Mosses

(c) Crustose lichens (d) Shrubs

2. Write short answers of the following questions.

i) Why biogeochemical cycle is named so?

ii) Differentiate between nitrification and de-nitrification?

iii) How does energy flow between trophic levels?

iv) What are the major producers in a terrestrial ecosystem?

v) How does ecological succession affect the community?

vi) What are the causes of ozone layer depletion?

vii) Why should we conserve biodiversity?

viii) Differentiate between renewable and non-renewable resources.

3. Write the details answers of the following questions:

i) What is a biogeochemical cycle? Explain nitrogen cycle in detail.

ii) Describe the effects of Carbon dioxide and global warming on human health.

iii) What is succession? Describe different stages of secondary succession with examples.

iv) What factors determine the growth rate of the human population? Give their reason also

EXERCISE

1. Encircle the correct choice.

i) Cloning of genes creates an identical copy of a DNA sequence. The process of cloning is significant for what reason?
(a) To create a transgenic organism
(b) To diagnose genetic disorders
(c) To study gene function
(d) All the above

ii) DNA sequencing is a technique used to determine the sequence of nucleotides in a DNA molecule. What is the significance of DNA sequencing?
(a) To identify genetic variations
(b) To create a transgenic organism
(c) To study gene expression
(d) All the above

iii) DNA analysis is used to identify genetic disorders. What is the purpose of DNA analysis?
(a) Sequence of nucleotides in a DNA
(b) To study gene expression
(c) To identify genetic variations
(d) All the above

iv) Biotechnology has many applications in healthcare, including gene therapy, drug production, and diagnosis of diseases. What is the significance of biotechnology in healthcare?
(a) To study gene function
(b) To improve animal welfare
(c) To develop new materials
(d) All the above

v) Polymerase chain reaction (PCR) is a technique used to amplify DNA. What is the significance of PCR?
(a) To sequence DNA
(b) To identify genetic disorders
(c) To study gene expression
(d) All the above

vi) Gel electrophoresis is a technique used to separate DNA fragments based on size. What is the significance of gel electrophoresis?
(a) To study gene expression
(b) To sequence DNA
(c) To identify genetic variations
(d) All the above

vii) Sanger sequencing is a technique used to read the sequence of DNA. What is the significance of Sanger sequencing?
(a) To identify genetic disorders
(b) To sequence DNA
(c) To study gene expression
(d) All the above

viii) Southern blotting is a technique used to transfer DNA fragments from a gel to a membrane. What is the significance of Southern blotting?
(a) To identify genetic disorders
(b) To sequence DNA
(c) To study gene expression
(d) All the above

ix) Genomic maps are useful for:
(a) Identifying genes associated with specific diseases
(b) Creating genetically modified organisms
(c) Analyzing DNA methylation patterns
(d) Detecting protein-protein interactions

x) What is the significance of monoclonal antibodies in biotechnology?
(a) To study gene expression
(b) Used to diagnose diseases
(c) Genetically modified organisms
(d) They can be used to create new drugs

2. Write short answer of the following:

i) Why Amp^R and Lac^Z are used in construction of rDNA?
ii) What do you mean by RFLPs?
iii) How is biotechnology used in healthcare, and what are some of the potential benefits and challenges associated with its use?
iv) Why restriction enzymes are called molecular scissors?

SUMMARY

- The effective and proper control of different, serious, vulnerable or common diseases of population by using various appropriate measures is called integrated disease management.
- Effective disease management can be achieved by implementing healthy habits and practices.
- The control of disease through other living organisms (predators) that reduces the severity of diseases.
- The initial vaccine was developed by Edward Jenner, a British physician, in 1796. Jenner employed the cowpox virus as a vaccine to safeguard against smallpox, a related virus affecting humans.
- The process of introducing vaccines into the body elicits protection against specific diseases and is known as vaccination.
- The Varicella vaccine is specifically designed to prevent this viral infection.
- Polio symptoms may include fever, reduced appetite, nausea, vomiting, sore throat, constipation, and abdominal pain.
- Animals play a vital role in providing a diverse range of highly nutritious food products, emphasizing the need for proper care and attention.

EXERCISE

1. Encircle the correct answer:

i) DT vaccine is used for
(a) Hepatitis (b) Measles
(c) Common cold (d) Diphtheria toxoid

ii) *Clostridium tetani* which caused tetanus is a
(a)Virus (b) Bacteria
(c) Parasite (d) Fungi

iii) MMR vaccine is effective for
(a) Measles (b) Polio
(c) Measles, Mumps & rubella (d) Hepatitis

iv) The top milk producing animal of Pakistan is
(a) Bhadawari (b) Jaffrabadi
(c) Nili- Ravi (d) Godavari

v) The yeast that gives puff appearance to dough is called
   (a) *Saccharomyces* (b) *Lactobacillus*
   (c) *Penicillium* (d) *Escherichia coli*

vi) The removal of floating solids and organic materials from the sewage is
   (a) Primary treatment (b) Secondary treatment
   (c) Tertiary treatment (d) none of these

vii) The process of hybrid crossing with its parents is called
   (a) Selection (b) Hybridization
   (c) Back cross (d) Acclimatiz ation

viii) French beans and Cauliflowers are vegetables of
   (a) Winter (b) Summer
   (c) Spring (d) all seasons

ix) The Science that deals with the modification of DNA and enhances the characteristic of an organism is called
   (a) Genetic Engineering (b) Biotechnology
   (c) Biology (d) None of these

x) Sludge digestion is a biological process in which organic solids are decomposed into stable substances
   (a) Stable substances (b) non stable substances
   (c) Effluent (d) None of these

2. Write short answer of the following

i) Why vaccination is important for infectious disease?
ii) List the outstanding milk producing breeds of cow.
iii) What do you mean by integrated diseases and its management?
iv) What is the role of livestock?
v) Differentiate between following
   (a) Selection and Backcross
   (b) Vegetables of summer and vegetables of winter
   (c) Selection and Hybridization

3. Give detail answer of the following questions

i) Explain in detail different steps of sewage treatment.
ii) Explain vaccination and its importance.
iii) Describes the role of microbes in food processing and sewage treatments.